

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

1. Given $(x_0, f_0), (x_1, f_1), \dots, (x_n, f_n)$, $x_i \neq x_j$ for $i \neq j$, let $p(x) = \sum_{i=0}^n f_i \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_i}{x_j - x_i}$.

Express the operational cost to evaluate p at some x as a function f of n .

Justify your formula for f explaining how you derived $f(n)$.

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Answer:

1. To derive f , we reason directly on the formulas:

- (a) Evaluating $\prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_i}{x_j - x_i}$ requires $2n$ subtractions (**sub**), n divisions (**div**), and n multiplications (**mul**).
- (b) Running this product $(n + 1)$ times: $2n(n + 1)$ **sub**, $n(n + 1)$ **div**, and $n(n + 1)$ **mul**.
- (c) Computing the sum $\sum_{i=0}^n f_i \star \Pi$ takes $n + 1$ additions (**add**) and $n + 1$ multiplications (**mul**).

So, $f(n) = 2n(n + 1)$ **sub** + $n(n + 1)$ **div** + $(n + 1)$ **add** + $(n(n + 1) + (n + 1))$ **mul**,

or summarizing: $f(n) = 4n^2 + 6n + 2$.